CCSA104

PRACTICAL ASSIGNMENT REPORT

Managing Security Policies

Student information	Details
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Course	CCSA104 – Security Policy Management
Practical environment	Check Point SmartConsole – A-SMS only
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1. Introduction

This report documents the practical work completed for the CCSA104 Security Policy Management assignment. The exercise focused on inspecting the Check Point policy-management interface, reviewing policy settings, creating network and host objects, building an Access Control rulebase, arranging rules into sections, configuring automatic Hide NAT, and reviewing policy installation history.

The practical scenario is Orion Logistics, a small organisation with an Internal LAN, a DMZ containing public-facing web and mail servers, and an External network. The intended policy follows the Positive Control Model: explicitly allow required traffic, deny other traffic, and log relevant decisions.

2. Lab environment and scope

The practical work was performed using the A-SMS (Security Management Server) environment through SmartConsole. The local lab did not include the Gateway Cluster or A-SME. The instructor confirmed that using A-SMS alone was acceptable for completing the project.

As a result, this report records configuration work performed in SmartConsole and the installation attempts visible in the supplied evidence. It does not claim successful policy enforcement or end-to-end traffic testing on a Security Gateway.

The assignment brief did not include Task 3, Task 5, or the definition of Test T4. In accordance with the instructor's direction, these activities were excluded. The placeholder rule labelled “Internal to DMZ” appears in the rulebase screenshot but is not treated as a completed rule because its required configuration was assigned to the omitted Task 5.

3. Practical work

3.1 Task 1 – Explore the default policy

SmartConsole was opened and the Security Policies workspace was inspected. The supplied screenshots show the policy-management interface and the available configuration areas. Global Properties was opened from the Applications menu to review the categories of global policy settings. The Manage Policies and Layers window was also opened to inspect the policy package list.

The available evidence includes the Global Properties window and the Manage Policies and Layers window. The screenshots supplied do not clearly establish every requested Cleanup rule property or the exact name of the active policy package. Those values should therefore be transcribed from SmartConsole if a fully populated Task 1 record is required; they have not been guessed in this report.

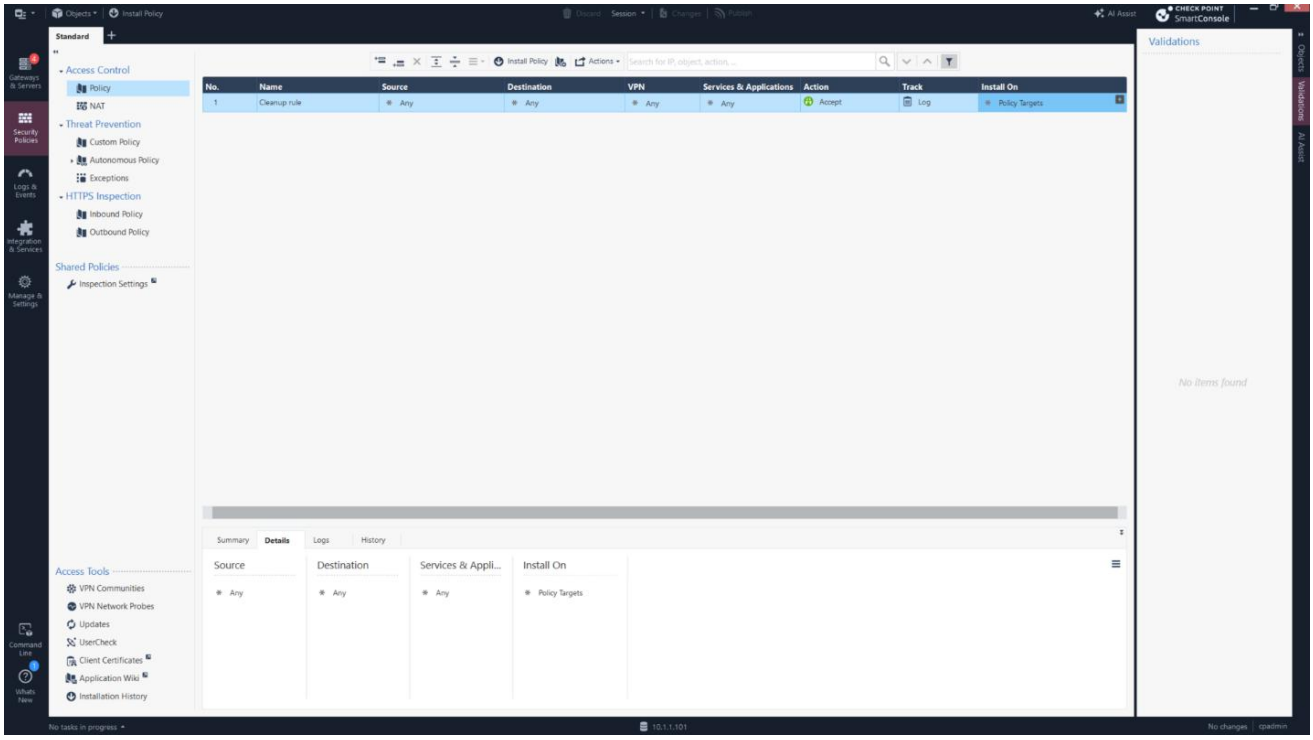


Figure 1. SmartConsole policy workspace inspected during Task 1.

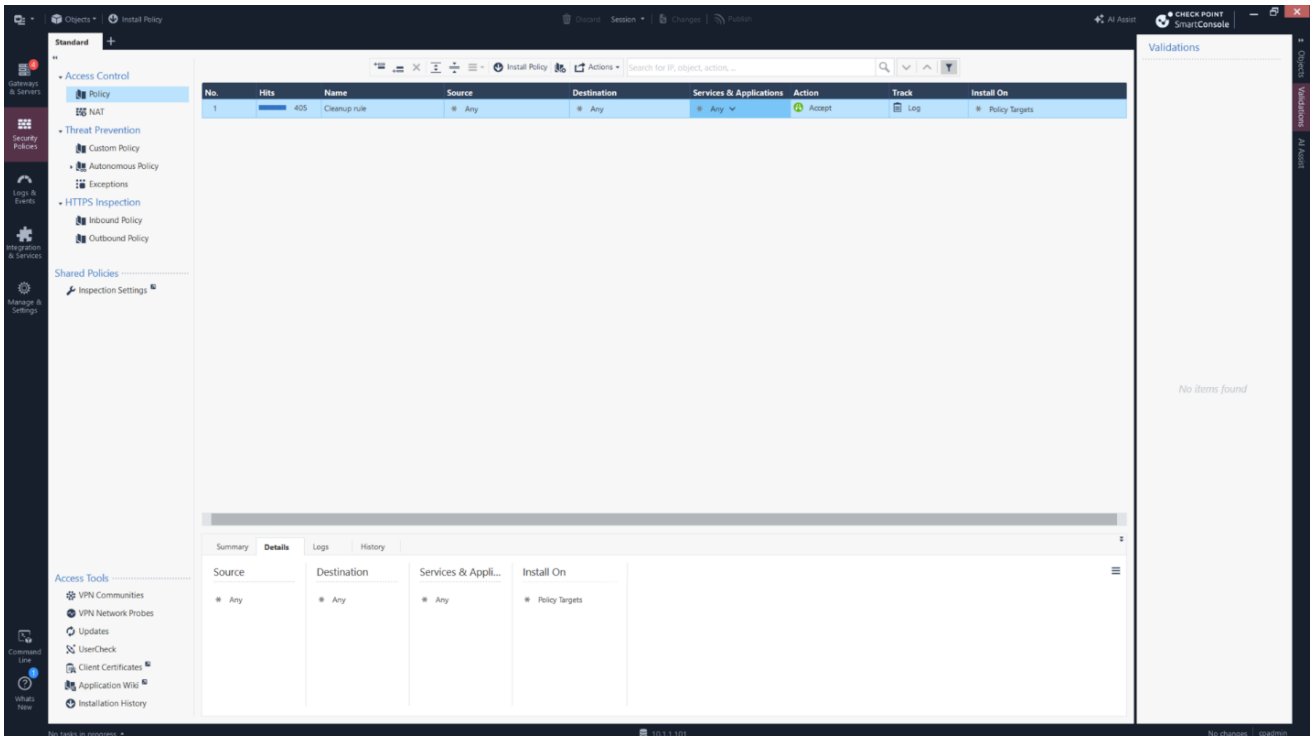


Figure 2. Policy workspace view supplied as Task 1 evidence.

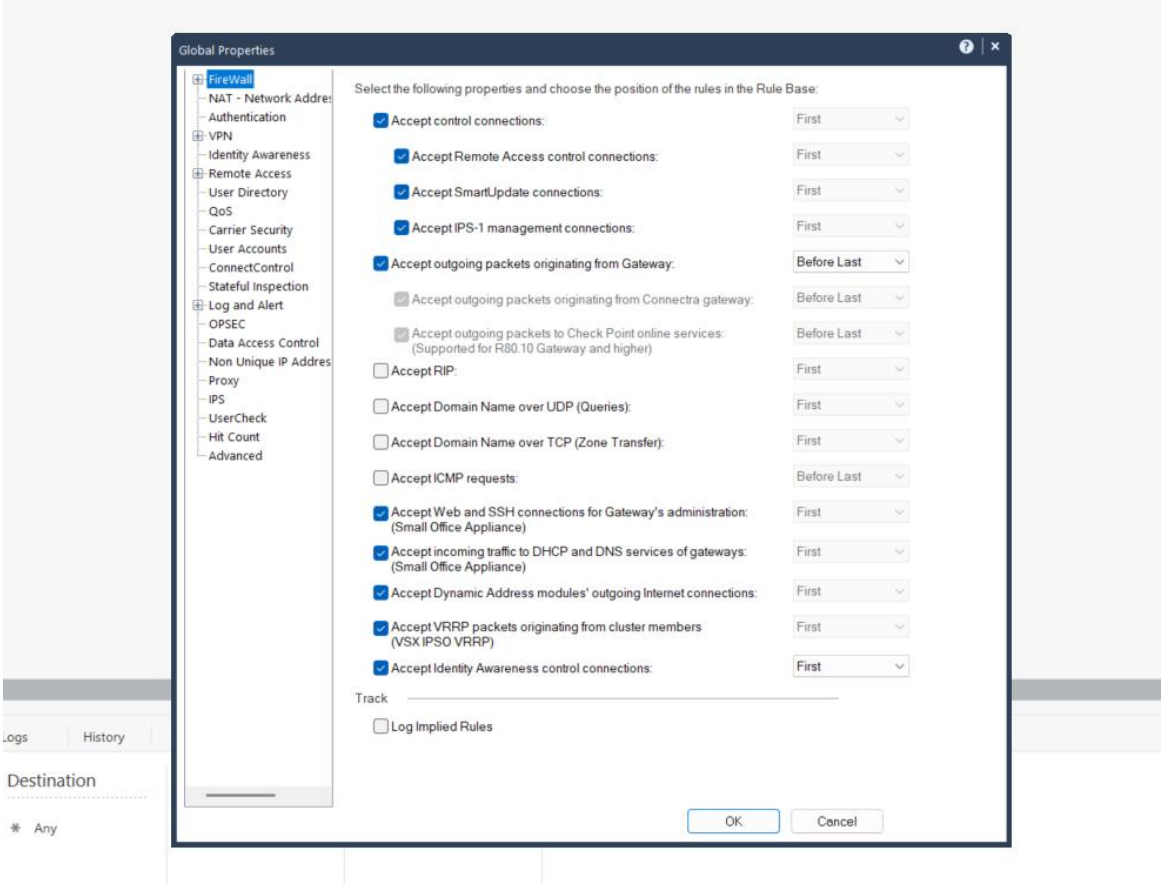


Figure 3. Global Properties window showing categories of global settings.

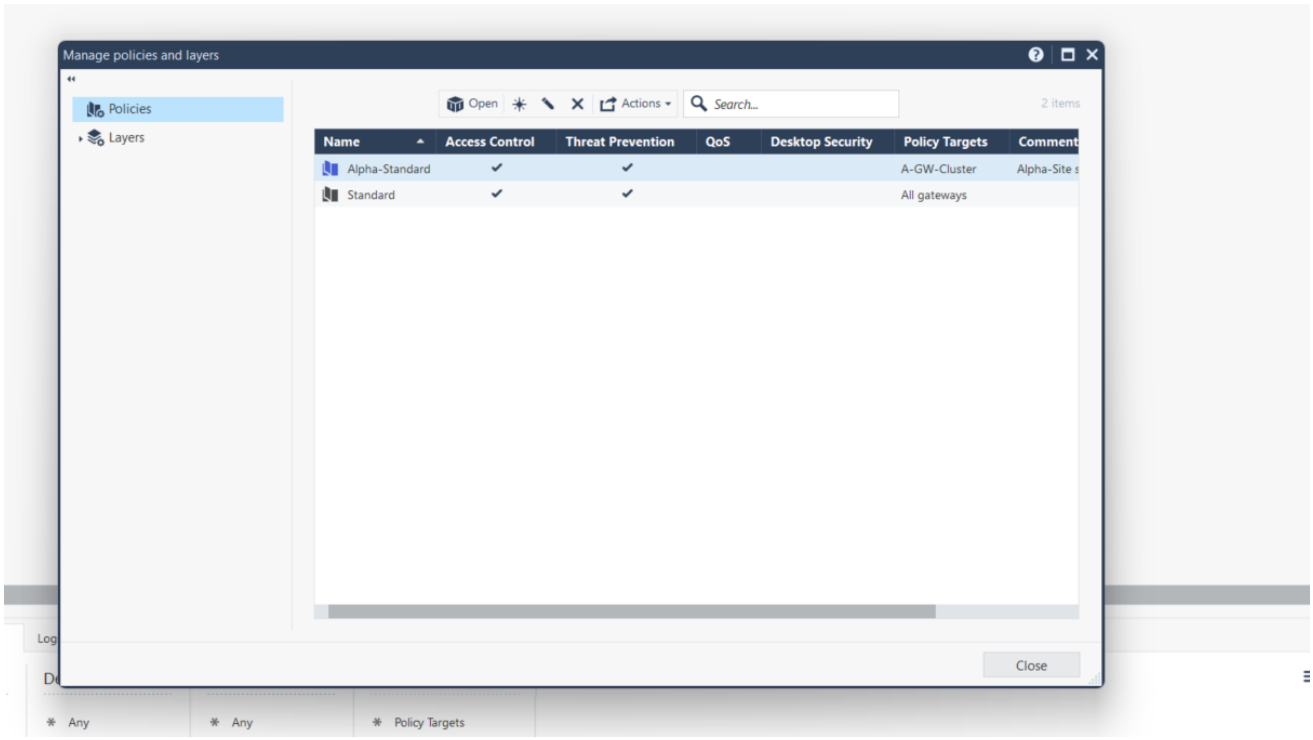


Figure 4. Manage Policies and Layers window showing available policy package information.

3.2 Task 2 – Create objects and build the Access Control rulebase

The required network and host objects were created or reviewed in SmartConsole. The assignment provides example addressing and instructs students to use the addressing in their own lab topology. The following inventory reflects the

example values specified in the assignment brief; where the screenshots show object names, those names are retained.

Object	Type	Example address	Purpose
Internal_LAN	Network	10.1.1.0/24	Staff network
DMZ_Net	Network	172.16.10.0/24	DMZ network
Admin_PC	Host	10.1.1.101	Administrator workstation
Web_Server	Host	172.16.10.20	Public web server
Mail_Server	Host	172.16.10.25	Public mail server
External_DNS	Host	192.9.100.10	External DNS server

The Access Control rulebase screenshot shows the configured rules and their visible Source, Destination, Services & Applications, Action, Track and Install On fields. The rules shown include Admin Access, Stealth Rule, Inbound to DMZ, DNS Out, Internal Web Out, SMTP Out and Cleanup rule. The rulebase also contains an “Internal to DMZ” placeholder with incomplete fields; because Task 5 was not supplied, no missing rule details have been inferred.

Rule	Source	Destination	Services	Action	Track
Admin Access	Admin_PC	A-GW-01 (gateway object shown)	ssh, https	Accept	Log
Stealth Rule	Any	A-GW-Cluster (object shown)	Any	Drop	Log
Inbound to DMZ	Any	Web_Server, Mail_Server	http, https, smtp	Accept	Log
Internal to DMZ	Not completed	Not completed	Not completed	Drop shown	Not set/shown
DNS Out	Internal_LAN	External_DNS	dns	Accept	Log
Internal Web Out	Internal_LAN	Any	http, https	Accept	Log
SMTP Out	Mail_Server	Any	smtp	Accept	Log
Cleanup rule	Any	Any	Any	Drop	Log

The screenshot displays A-GW-01 for Admin Access and A-GW-Cluster for the Stealth Rule. These object labels are reported as shown in the evidence; they are not interpreted as proof that a gateway cluster was installed or available for policy enforcement.

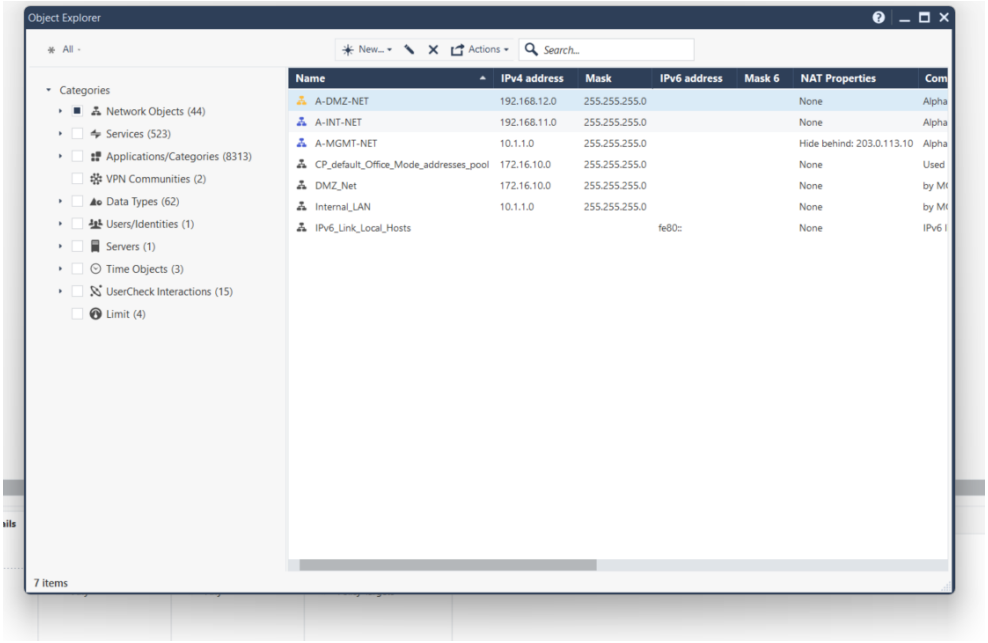


Figure 5. Objects and network configuration visible in SmartConsole.

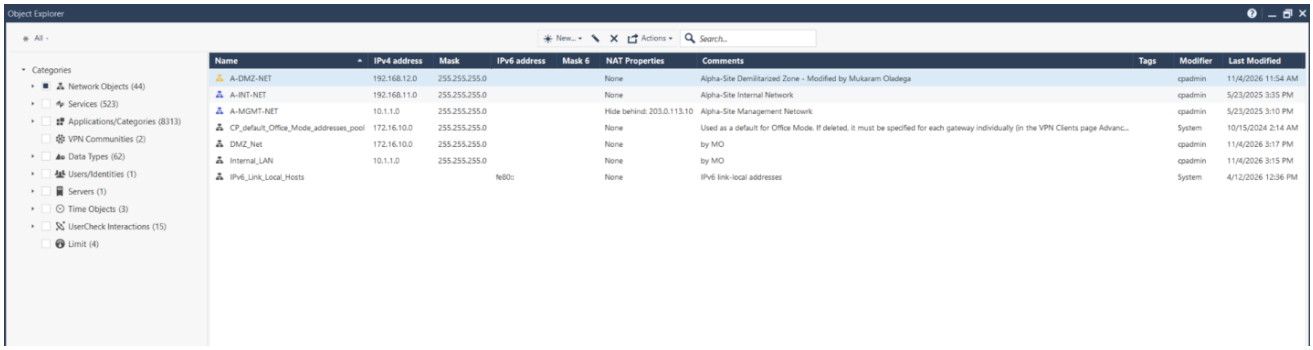


Figure 6. Object inventory displayed in SmartConsole.

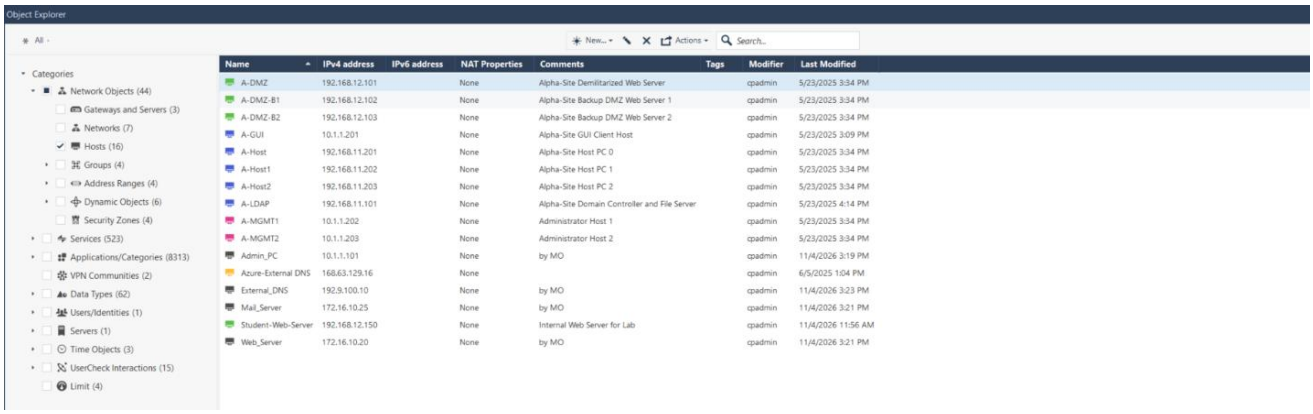


Figure 7. Additional object list and object-management evidence.

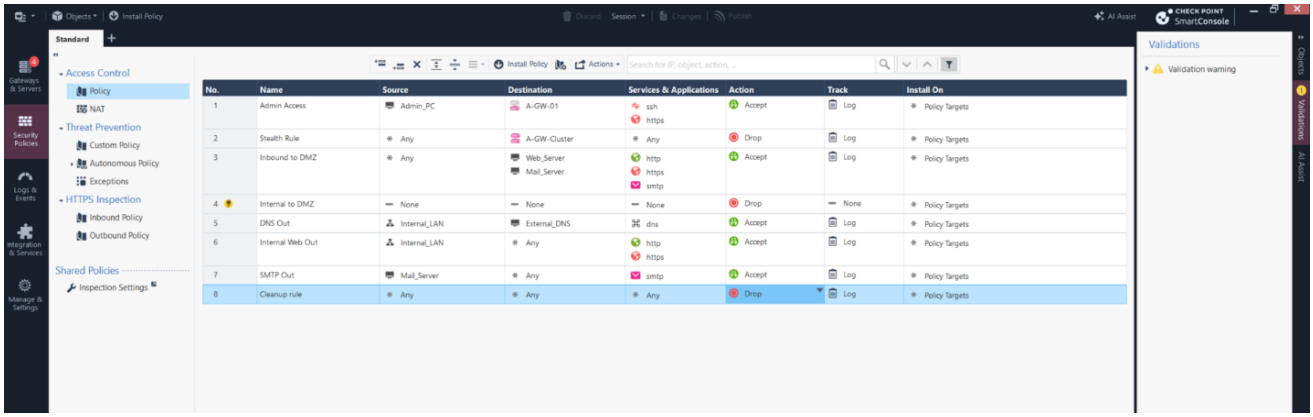


Figure 8. Access Control rulebase showing the configured rules and the incomplete Internal to DMZ placeholder.

3.3 Task 4 – Organise the rulebase with sections

The Access Control rulebase was organised using section titles to make the policy easier to read and manage. The screenshot evidence shows the rulebase grouped under section headings. The intended organisation in the assignment is:

Section title	Rules grouped in the section
Management and Protection	Admin Access and Stealth Rule
DMZ Services	Inbound to DMZ and the Internal to DMZ placeholder (incomplete)
Outbound Access	DNS Out, Internal Web Out and SMTP Out
Cleanup	Cleanup rule

Sections are organisational headings in the rulebase. They do not independently change rule enforcement order and are not installed as enforcement rules on the Security Gateway. Enforcement continues to depend on the order of the actual rules.

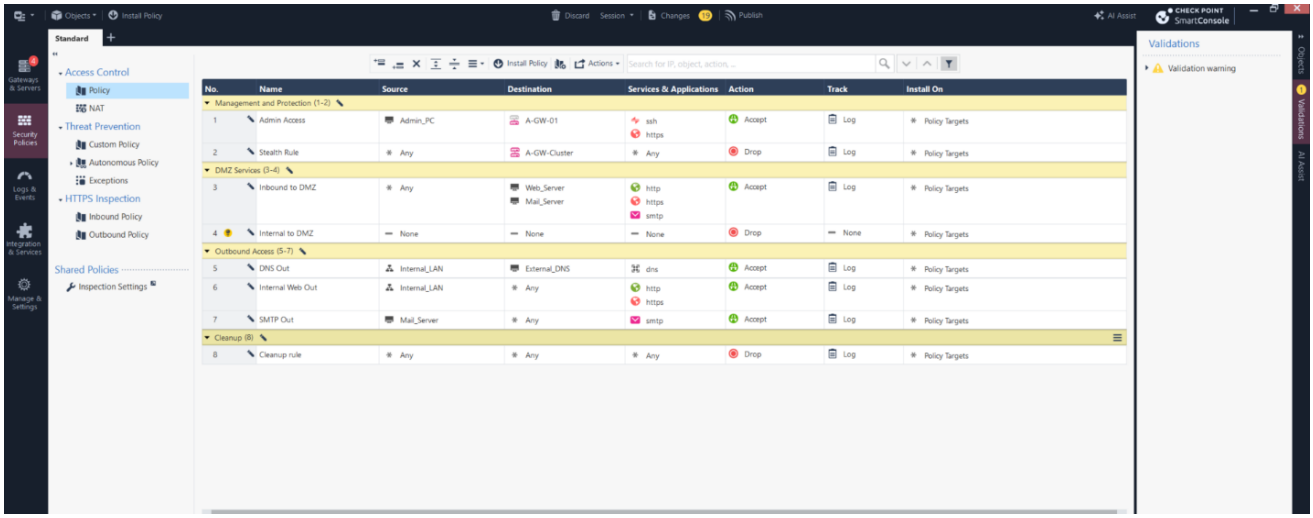


Figure 9. Rulebase displayed with section headings and configured rules.

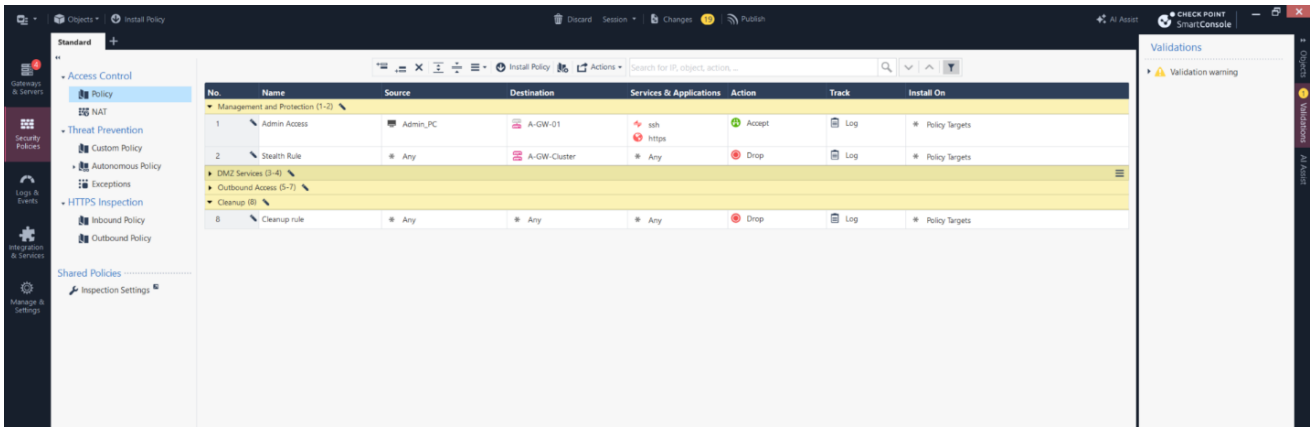


Figure 10. Further view of the section-organised rulebase.

3.4 Task 6 – Configure automatic Hide NAT

The Internal_LAN network object was opened and its NAT settings were inspected. The supplied screenshot shows automatic address translation enabled, with Hide selected and the network configured to be hidden behind the gateway. This is an object-based automatic NAT configuration rather than a separately authored manual NAT rule.

Hide NAT provides many-to-one translation: multiple internal hosts can share a translated public or gateway address when communicating externally. The gateway uses source-port translation (PAT) to keep concurrent connections distinguishable and to map returning traffic to the correct internal session. The screenshots also show the NAT rulebase containing automatically generated entries.

The configuration is documented as completed in the management interface. Because the lab did not include an available gateway for end-to-end testing, successful Internet access through this NAT configuration was not verified.

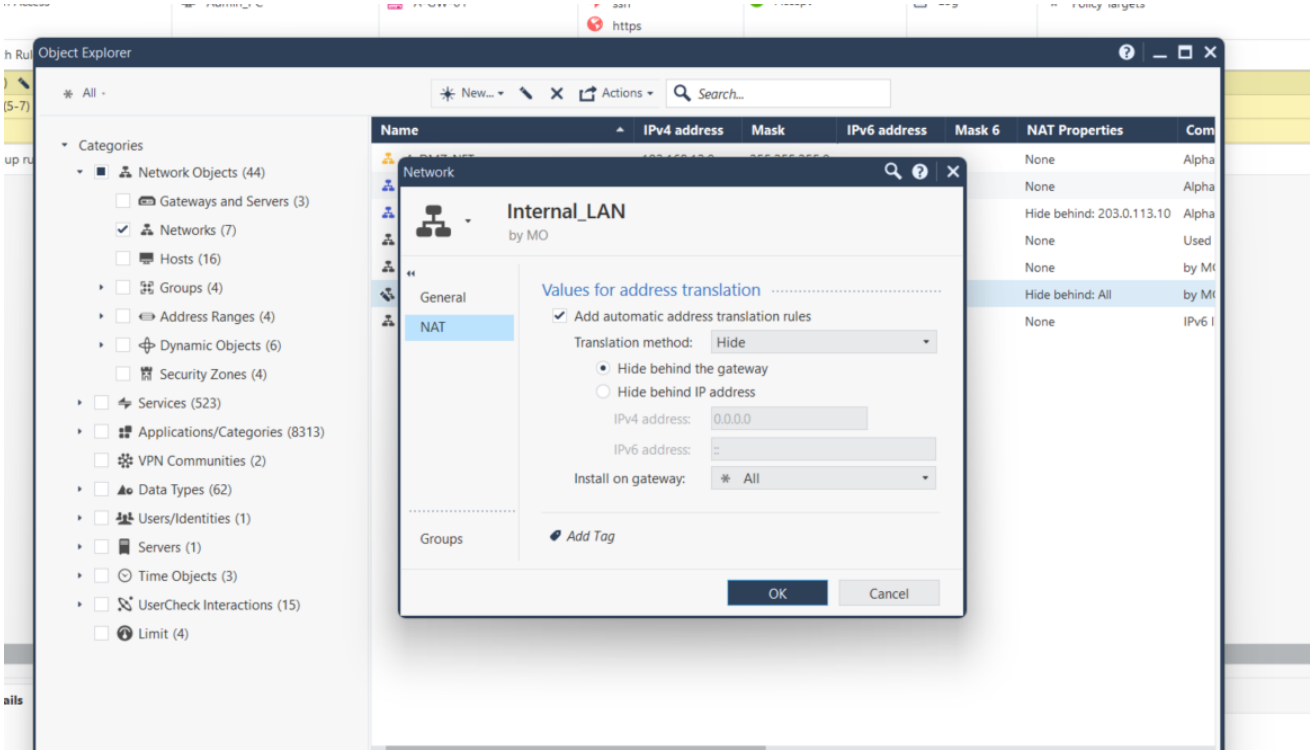


Figure 11. Internal_LAN object NAT settings showing automatic Hide NAT configuration.

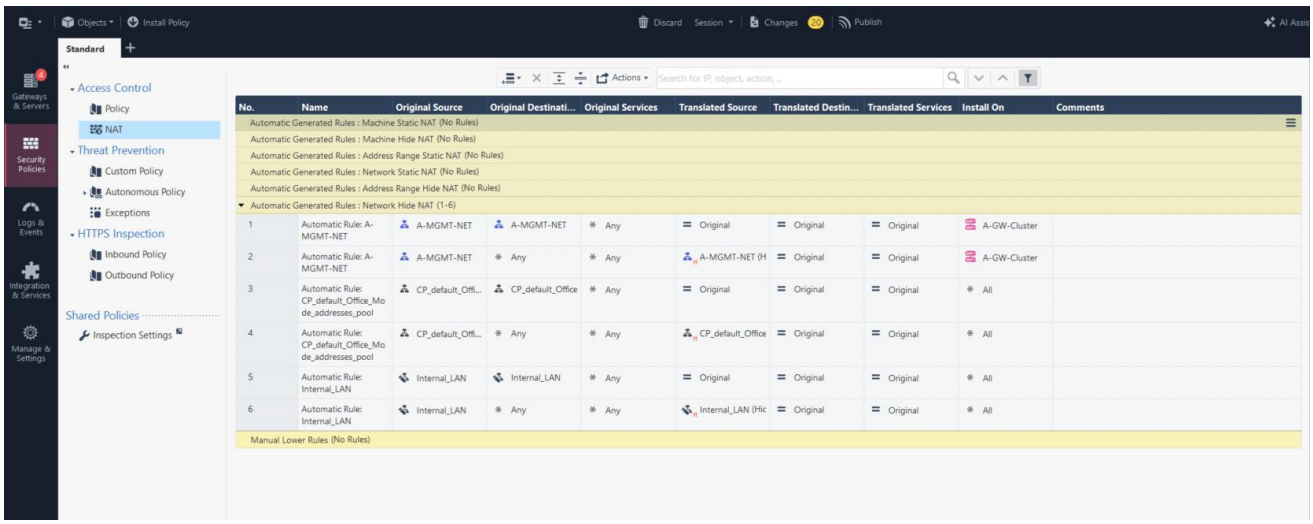


Figure 12. NAT rulebase showing the automatically generated translation rule entries.

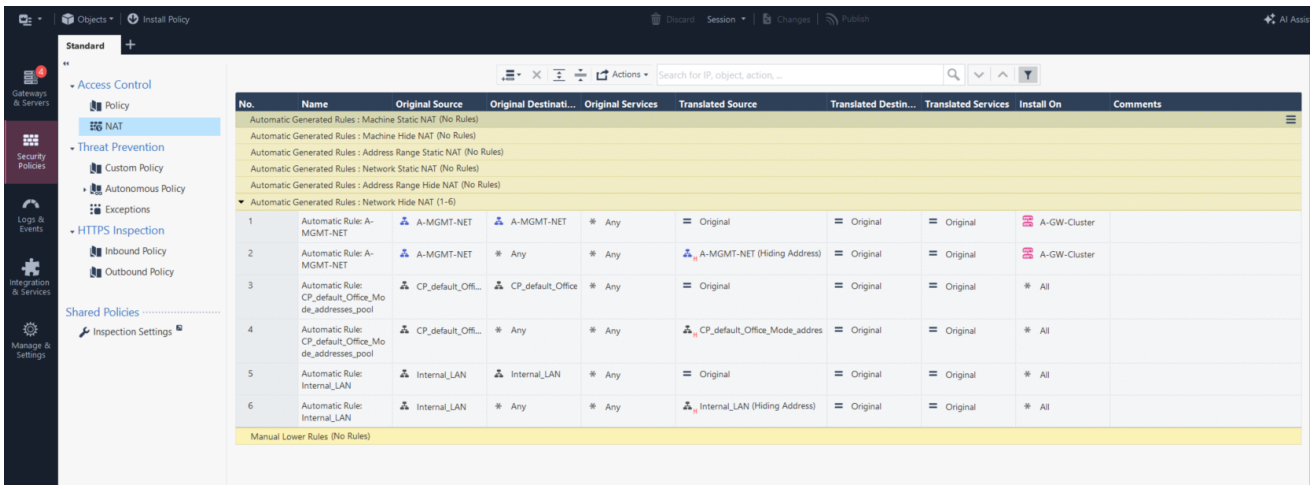


Figure 13. Additional NAT rulebase view supplied as evidence.

3.5 Task 7 – Rule-order experiment and Installation History

The assignment asks students to insert a temporary rule named TEMP Block Internal above Internal Web Out, publish and install the policy, repeat Test T4, move the temporary rule below Internal Web Out, repeat the test, and then delete the temporary rule. However, Test T4 was not provided by the instructor and the gateway cluster/A-SME were not available in the local environment. Therefore, traffic-match results for T4 cannot be reported.

The supplied screenshots show the rulebase with temporary rule/order changes and the Installation History area. They also show failed installation attempts. This is consistent with the stated lab limitation: A-SMS was available, but the Gateway Cluster and A-SME required for enforcement/testing were not installed locally. The report therefore records the visible installation result as a failure and does not describe the policy as successfully installed.

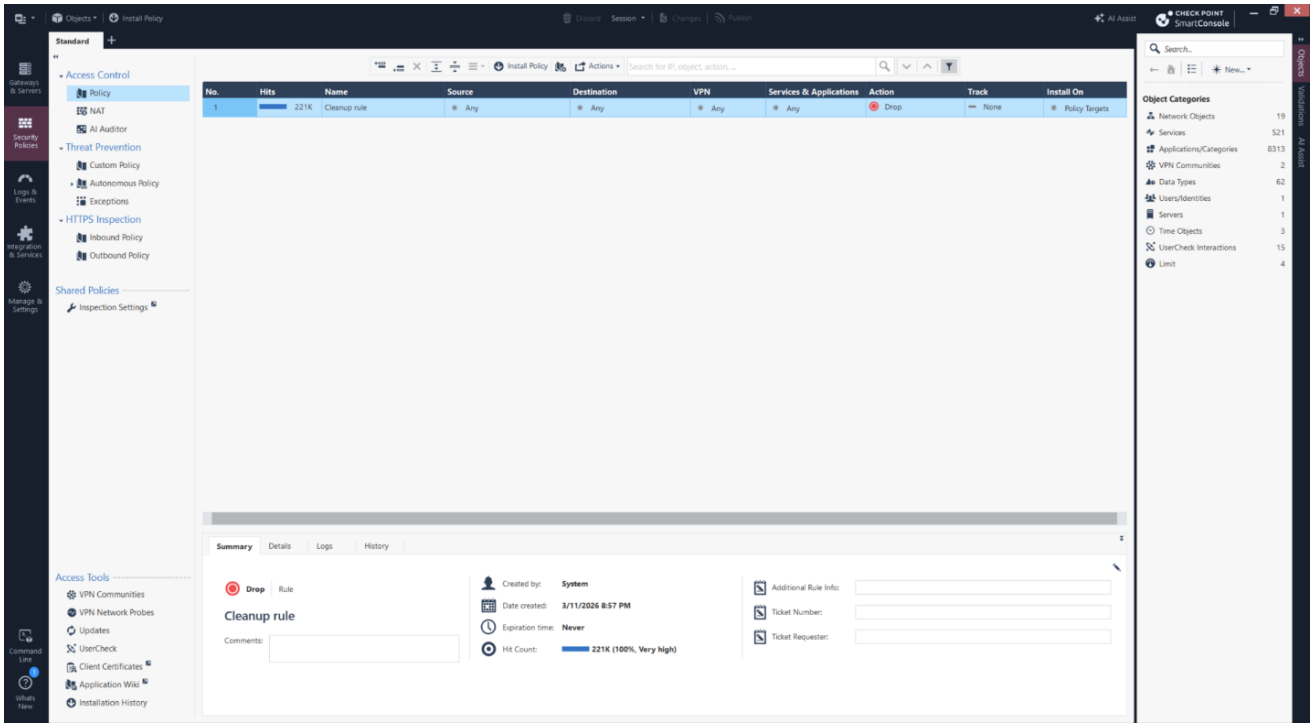


Figure 14. SmartConsole policy view associated with the installation and change workflow.

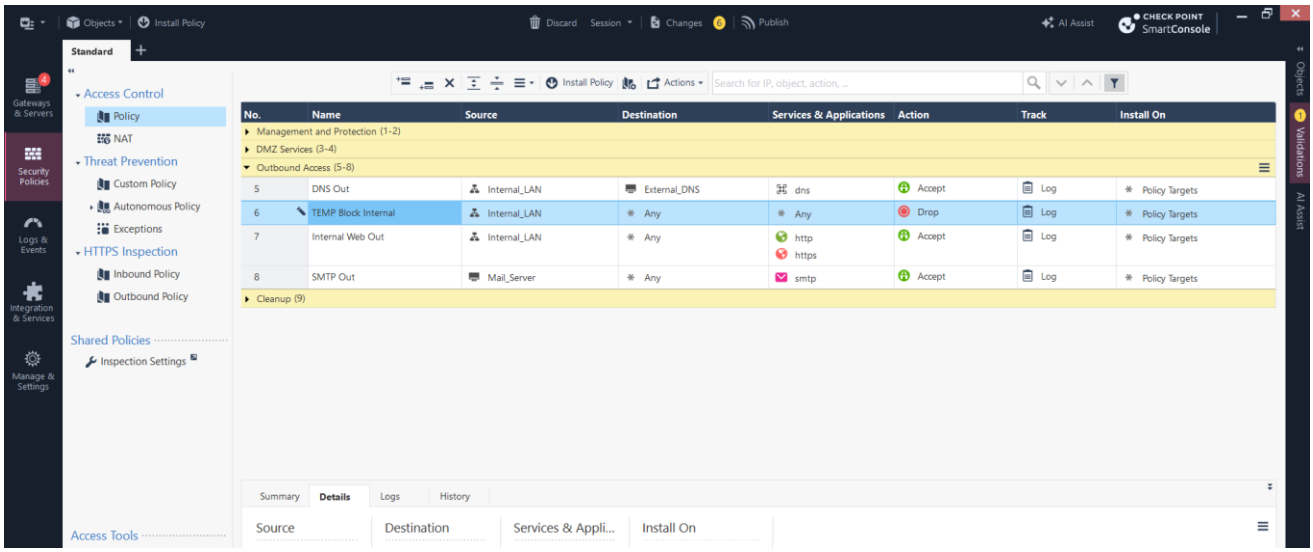


Figure 15. Installation History view showing installation records.

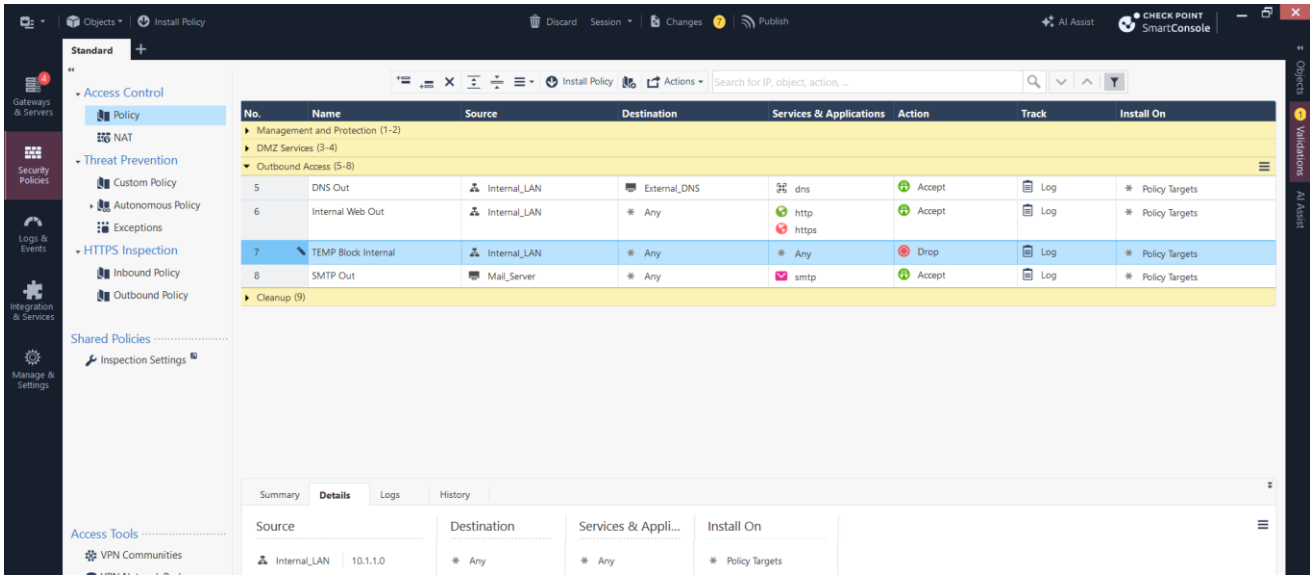


Figure 16. Installation History details and status information.

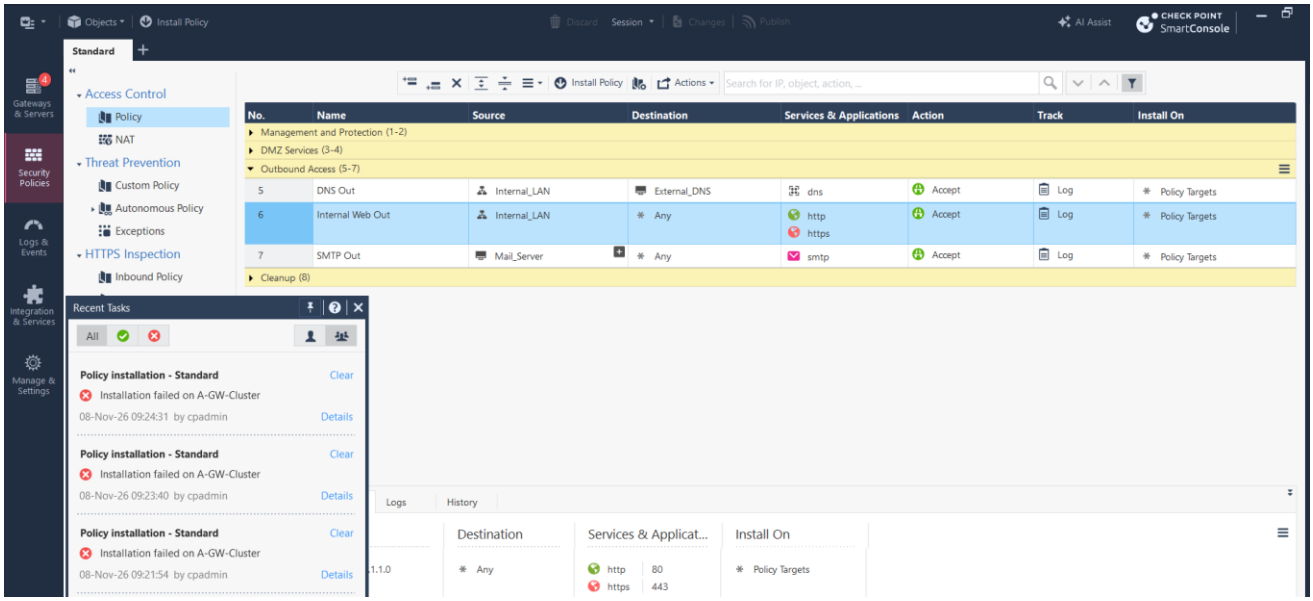


Figure 17. Installation record detail showing the failed installation result.

4. Discussion

The practical work demonstrates the management-side activities involved in constructing and organising a Check Point Access Control policy. The rulebase follows the Positive Control approach by defining specific permitted services and retaining a Drop action in the Cleanup rule for traffic that does not match an earlier rule.

Rule order remains important because the firewall evaluates rules in sequence. A specific rule placed above a broader rule can permit or deny the intended traffic before a later rule is evaluated. The Stealth Rule is intended to protect the gateway itself, while the Cleanup rule remains at the bottom to drop and log unmatched traffic. The section headings improve readability but do not replace correct rule ordering.

The Hide NAT configuration illustrates how a network object can automatically generate NAT rules. The management-side configuration can be reviewed in SmartConsole, but operational verification requires a functioning Security Gateway and suitable test hosts. That verification was outside the available lab environment.

5. Limitations and excluded activities

- A-SMS was the only Check Point component available locally. The Gateway Cluster and A-SME were not installed.
- The instructor approved completing the project using A-SMS only.
- Policy installation attempts shown in the screenshots failed; successful installation and gateway enforcement are not claimed.
- Task 3, Task 5 and Test T4 were omitted by the instructor and are excluded from this report.
- The Internal to DMZ rule is visible as an incomplete placeholder. Its configuration was not invented because the assignment says it is added in the omitted Task 5.
- End-to-end traffic testing, NAT connectivity testing and T4 log-match verification could not be established.

6. Conclusion

The practical assignment activities that were possible in the A-SMS-only environment were carried out and documented. The evidence covers SmartConsole policy inspection, object management, Access Control rule configuration, section organisation, automatic Hide NAT settings, NAT rulebase review and Installation History inspection. The main limitation was the absence of a local Security Gateway/A-SME, which prevented successful policy installation and traffic-level verification. The report distinguishes completed management configuration from tasks that required the unavailable gateway environment or were excluded by the instructor.

Appendix A – Evidence index

Figure	Evidence description
1–2	SmartConsole policy workspace
3	Global Properties
4	Manage Policies and Layers
5–7	Network and host object views
8	Access Control rulebase
9–10	Rulebase sections
11	Internal_LAN automatic Hide NAT settings
12–13	NAT rulebase
14–17	Policy workflow and Installation History

7. Part B – Theory Questions (30 marks)

The following answers address all twelve theory questions in the CCSA104 assignment. They explain the concepts in the context of the Orion Logistics policy and the Check Point SmartConsole workflow.

Section B1 – Multiple-choice questions (5 marks)

Question	Correct option	Answer
Q1	B	Created in SmartConsole, stored on the Security Management Server, and enforced by Security Gateways.
Q2	A	Positive Control Model (whitelist).
Q3	C	As early as possible, typically immediately after the Management rules.
Q4	D	Sections are visual divisions only and do not affect enforcement order.
Q5	B	Many-to-one translation.

Section B2 – Short-answer questions (15 marks)

Q6. Distinguish between Explicit rules and Implied rules. State where Implied rules are configured and where they can be placed. What is the Implicit Cleanup Action?

Explicit rules are rules that an administrator deliberately creates in the Access Control rulebase. The administrator specifies their source, destination, services, action, tracking and position. For example, the Orion Logistics Admin Access rule explicitly permits SSH and HTTPS management traffic from Admin_PC to the gateway.

Implied rules are system-generated rules that Check Point adds to support required firewall and management functions. They are not created as ordinary manually authored rules in the Access Control rulebase. Their behaviour is controlled through Global Properties, under the relevant Firewall/Implied Rules settings. The available placement options determine where enabled implied rules are evaluated in relation to the explicit rulebase; depending on the setting and rule type, they can be placed before or after the applicable explicit rules. Administrators should review these settings because an implied rule can affect which traffic is accepted or dropped.

The Implicit Cleanup Action is the default action applied to traffic that does not match an applicable explicit or implied rule. In a Positive Control policy, the intended default is Drop, so unapproved traffic is denied rather than being allowed automatically. Logging should be enabled where the policy and platform settings support it so unmatched traffic can be investigated.

Q7. Explain why rule order matters in a rulebase. Give one example from your own rulebase.

A firewall evaluates rules from the top of the rulebase downwards and applies the first rule that matches the traffic. Therefore, a broad rule placed too high can match traffic before a more specific rule is reached, making the later rule ineffective for that traffic. Rule order is important for preserving the intended security controls and ensuring that specific exceptions are processed before general rules.

For example, the Admin Access rule should appear before the Stealth Rule so that the approved administrator workstation can reach the gateway using SSH and HTTPS. The Stealth Rule then drops other traffic addressed to the gateway. Similarly, a temporary Drop rule for Internal_LAN placed above Internal Web Out would match internal web traffic first and block it; placing it below Internal Web Out allows the web rule to match first.

Q8. Explain the difference between saving and publishing changes in SmartConsole. What happens to unpublished changes when you install a policy, and why do many organisations publish only during a change window?

Saving preserves the administrator's current work in the SmartConsole session so it can be continued later. Publishing commits the session changes to the Security Management Server, making the changes part of the management database and available for policy installation. Publishing is therefore the formal commit step, while saving alone does not make the changes part of the published policy.

Unpublished changes are not included in the policy version that is installed on the Security Gateway. The gateway receives the published policy version selected for installation, not an uncommitted private session. Administrators must publish the intended changes before installing them.

Organisations commonly publish during an approved change window so changes are reviewed, coordinated and introduced at a controlled time. This reduces the risk of unexpected service disruption, conflicting administrator changes, accidental exposure, or changes being deployed during critical business operations. It also allows monitoring and rollback planning to be coordinated.

Q9. Compare Global Properties with Policy Packages. Give one example of what each controls and say how you open each in SmartConsole.

Global Properties contain system-wide or management-level settings that influence how Check Point gateways and policies behave. An example is the configuration of implied firewall rules or other global firewall behaviour. Global Properties can be opened from the Applications menu in SmartConsole.

A Policy Package is the container for the security policy components that an administrator manages and installs to selected Security Gateways. It can include Access Control and NAT policy layers, together with the installation targets. For example, the Orion Logistics Access Control rules and the associated NAT rules are managed as part of a policy package. Policy Packages and Layers can be opened from the Security Policies workspace using Manage Policies and Layers.

In short, Global Properties configure broader Check Point behaviour, while a Policy Package contains the actual policy rules and layers that are installed on selected gateways.

Q10. Name the four predefined Security Zones. State two rules about assigning gateway interfaces to zones, and one benefit of using zone-based rules.

The four predefined Security Zones are Internal, External, DMZ and Undefined.

Two important interface-assignment principles are:

- A Security Zone is associated with a gateway interface; traffic is classified into a zone according to the interface through which it enters or leaves the gateway. Assign the correct zone to each relevant gateway interface based on the network it connects to.
- An interface should have one clear zone assignment at a time. Do not attempt to classify the same interface as belonging to multiple zones; if the network design changes, review the interface-to-zone mapping and the rules that use those zones.

A benefit of zone-based rules is simpler and more readable policy administration. Instead of repeatedly listing individual network objects, an administrator can write rules between trust zones—for example, allowing only required traffic from Internal to DMZ. This also makes it easier to apply consistent controls when networks or hosts change within a zone.

Section B3 – Scenario questions (10 marks)

Q11. A new administrator inserts a rule with Source Any, Destination Any, Services Any and Action Accept above the Stealth rule. Describe two security problems this causes, explain how to correct it, and name the type of rule that is allowed to sit above the Stealth rule and why.

First, the rule is excessively permissive. It allows traffic from any source to any destination using any service, which defeats the Positive Control Model and violates the principle of least privilege. It could permit unnecessary or malicious traffic that should have been denied.

Second, because the rule is above the Stealth Rule and matches such a broad range of traffic, it can match traffic destined for the gateway before the Stealth Rule is evaluated. This undermines the protection that the Stealth Rule is intended to provide and may expose gateway services to unauthorised access.

The administrator should remove the broad Accept rule and replace it, if there is a legitimate requirement, with a narrowly scoped rule specifying the approved source, gateway destination, required management services and logging. The Management rules are the rules that may sit above the Stealth Rule. They are placed there to permit authorised administrators to manage the gateway; the Stealth Rule should follow them as early as possible to drop other traffic directed at the gateway.

Q12. After Hide NAT is configured for Internal_LAN, staff report they cannot reach the Internet.

(a) Two benefits of NAT:

- Public IPv4 address conservation: Hide NAT allows multiple internal devices to share one translated external address instead of requiring a separate public IPv4 address for every workstation.
- Address and topology concealment: external systems see the translated address rather than the individual private addresses of internal hosts, reducing direct exposure of the internal addressing scheme.

(b) Three checks to identify the fault:

Check	What to verify	Why it matters
1. Internal_LAN NAT object settings	Confirm automatic address translation is enabled, the translation method is Hide, and the network is hidden behind the correct gateway address/interface.	An incorrect object, translation method or translated address can prevent internal source addresses from being translated correctly.
2. NAT rulebase and policy installation	Confirm the automatically generated Hide NAT rule exists, applies to Internal_LAN, is not shadowed by an incorrect earlier NAT rule, and that the intended published policy has been installed on the gateway.	A correct object setting is not sufficient if the generated rule is missing, overridden, or not present on the gateway.
3. Routing, security policy and DNS	Check that the gateway has a valid route to the Internet and a return route where required; confirm the Access Control policy permits the necessary outbound services and that DNS resolution works.	NAT does not create Internet connectivity by itself. Traffic still needs a valid route and an allowing security rule; DNS failure can also make Internet access appear unavailable.

In a live environment, I would also review the gateway logs for the affected source host and test connectivity by IP address and by hostname. This helps distinguish a NAT or routing problem from a DNS-resolution problem. These operational tests were not possible in my A-SMS-only lab because no local Security Gateway was available.

8. Overall conclusion

The practical and theory components together demonstrate the key principles of Check Point Security Policy Management. The practical work covered SmartConsole policy inspection, object and rulebase management, section organisation, automatic Hide NAT configuration and Installation History review. The theory answers explain the Positive Control Model, rule evaluation order, explicit and implied rules, policy publishing, Global Properties, Policy Packages, Security Zones and NAT troubleshooting.

The practical evidence must be interpreted within the approved A-SMS-only lab scope. The instructor permitted this environment, but the absence of the Gateway Cluster and A-SME prevented successful gateway installation and live traffic verification. Task 3, Task 5 and Test T4 were also excluded by the instructor. Accordingly, this report documents only the activities and results supported by the available screenshots and does not claim that excluded or unavailable tests were completed.